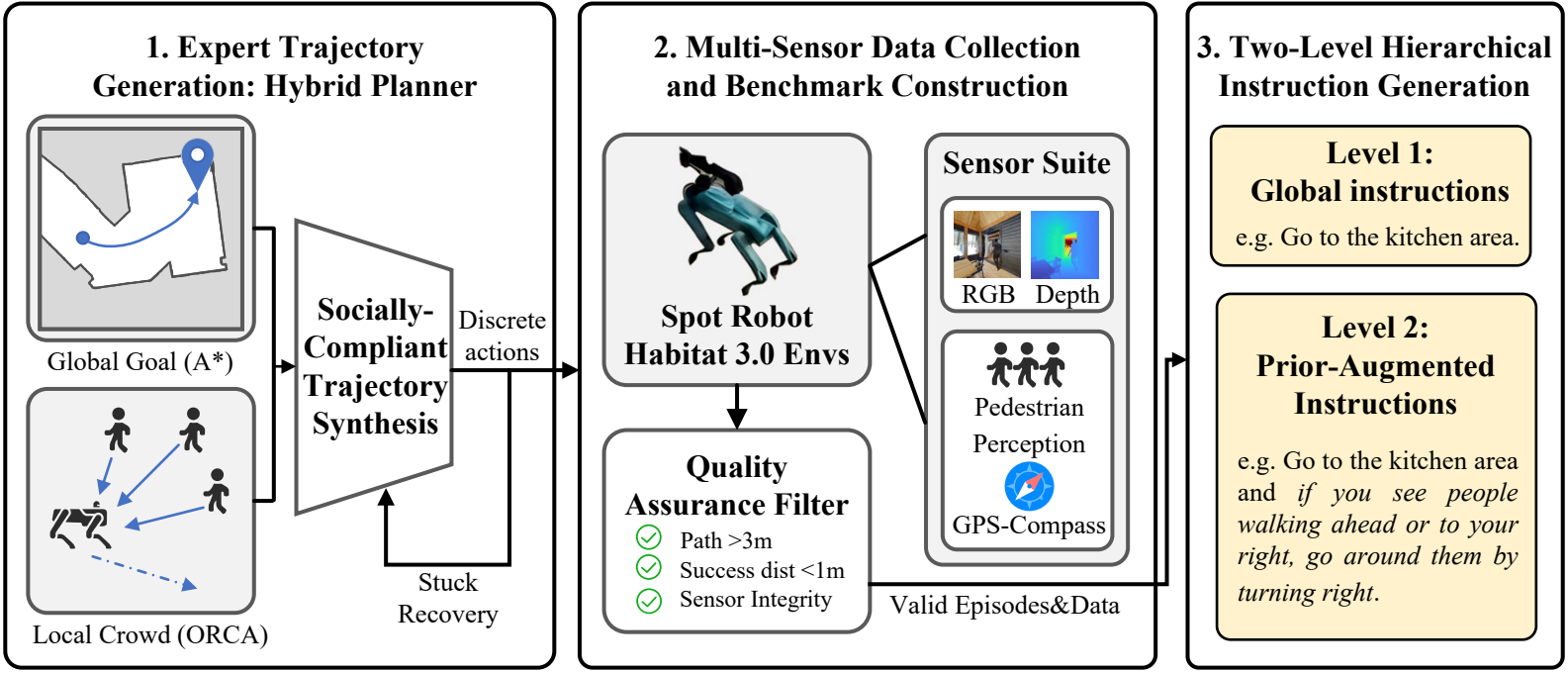

1. Expert Trajectory Generation: Hybrid Planner
Global Goal (A*)
Local Crowd (ORCA)
Socially-Compliant Trajectory Synthesis
Discrete actions
Stuck Recovery
2. Multi-Sensor Data Collection and Benchmark Construction
Spot Robot Habitat 3.0 Envs
Sensor Suite
RGB
Depth
Pedestrian Perception
GPS-Compass
Quality Assurance Filter
Path >3m
Success dist <1m
Sensor Integrity
Valid Episodes&Data
3. Two-Level Hierarchical Instruction Generation
Level 1: Global instructions
e.g. Go to the kitchen area.
Level 2: Prior-Augmented Instructions
e.g. Go to the kitchen area and if you see people walking ahead or to your right, go around them by turning right.